ORIGINAL RESEARCH

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Cost-Effectiveness of Extending Human Papillomavirus Vaccination to Population Subgroups Older Than 26 Years Who Are at Higher Risk for Human Papillomavirus Infection in the United States

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Background: In June 2019, the U.S. Advisory Committee on Immunization Practices recommended shared clinical decision making regarding potential human papillomavirus (HPV) vaccination of men and women aged 27 to 45 years ("mid-adults").

Objective: To examine the incremental cost-effectiveness ratios (ICERs) and number needed to vaccinate (NNV) to prevent 1 HPV-related cancer case of expanding HPV vaccination to subgroups of mid-adults at increased risk for HPV-related diseases in the United States.

Design: Individual-based transmission dynamic modeling of HPV transmission and associated diseases using HPV-ADVISE (Agent-based Dynamic model for VaccInation and Screening Evaluation).

Data Sources: Published data.

Target Population: All U.S. mid-adults and higher-risk subgroups within this population.

Time Horizon: 100 years.

Perspective: Health care sector.

Intervention: Expanding 9-valent HPV vaccination to mid-adults and higher-risk subgroups.

Outcome Measures: ICERs and NNVs.

Results of Base-Case Analysis: Expanding 9-valent HPV vaccination to all mid-adults, those with more lifetime partners, and those who have just separated was projected to cost an additional \$2 005 000, \$763 000, and \$1 164 000 per quality-adjusted life-year (QALY) gained, respectively. The NNVs to prevent 1 additional HPV-related cancer case were 7670, 3190, and 5150, respectively, compared with 223 for vaccination of persons aged 9 to 26 years (vs. no vaccination).

Results of Sensitivity Analysis: The mid-adult strategy with the lowest ICER and NNV was vaccinating infrequently screened mid-adult women who have just separated and have a higher number of lifetime sex partners (ICER, \$86 000 per QALY gained; NNV, 470).

Limitation: Uncertainty about rate of new sex partners and natural history of HPV among mid-adults.

Conclusion: Vaccination of mid-adults against HPV is substantially less cost-effective and produces higher NNVs than vaccination of persons younger than 26 years under all scenarios investigated. However, cost-effectiveness and NNV are projected to improve when higher-risk mid-adult subgroups are vaccinated, such as mid-adults with more sex partners and who have recently separated, and women who are underscreened.

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In 2006, the U.S. Advisory Committee on Immunization Practices recommended routine human papillomavirus (HPV) vaccination for females aged 11 and 12 years (series can be started at age 9 years) with catch-up vaccination through age 26 years (1). The recommendation was extended 5 years later to males aged 11 and 12 years (series can be started at age 9 years) with catch-up through age 21 years (2). In June 2019, following the U.S. Food and Drug Administration's expanded approval of the 9-valent HPV vaccine for use through age 45 years (3) and after reviewing evidence related to HPV vaccination of adults, the Committee recommended catch-up HPV vaccination for all females and males through age 26 years and shared clinical decision making for HPV vaccination of women and men aged 27 through 45 years ("mid-adults") (4). Evidence reviewed by the Committee included mathematical models from 5 groups that suggested that, although vaccination is safe, the population-level benefit of vaccinating mid-adults would be small (5). However, at the individual level, mid-adults who are at higher risk for acquiring new infections (for example, adults with multiple new partners or who are separated after long-term relationships) or for developing HPV-related cancer (for example, women who participate less in cervical cancer screening) could benefit

See also:

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substantially from HPV vaccination (6). In this context, more information on the cost-effectiveness of vaccinating mid-adult subgroups at higher risk for HPV infection and the number needed to vaccinate (NNV) to prevent 1 additional HPV-related cancer case can help inform shared clinical decision making about potential HPV vaccination. The objective of the current study was to evaluate the incremental cost-effectiveness and NNV of extending HPV vaccination to subgroups of the mid-adult U.S. population that may be at higher risk for acquiring HPV infection.

METHODS

Model Overview

For model simulations, we used the U.S. version of HPV-ADVISE (Agent-based Dynamic model for VaccInation and Screening Evaluation) (7, 8), a calibrated, individual-based, transmission dynamic model that integrates the following 6 components: 1) demographic characteristics (open stable population aged 10 to 100 years), 2) sexual behavior and HPV transmission (18 HPV types modeled separately, including vaccine types 6, 11, 16, 18, 31, 33, 45, 52, and 58), 3) type-specific natural history of disease (infection; anogenital warts; precancerous cervical lesions [3 grades]; cancer of the cervix [3 stages]; and HPV-attributable cancer of the anus, oropharynx, penis, vagina, or vulva), 4) vaccination, 5) screening and treatment, and 6) health economic outcomes. The behavioral and biological parameter values were derived by calibrating (fitting) the model to highly stratified U.S. data on sexual behavior, HPV epidemiology, and screening taken from the literature and population-based data sets (7, 8). For model projections, we identified the 50 best-fitting parameter sets using data targets and least squares (see the Technical Appendix [8]). External predictive validation of the model has been carried out by comparing model projections versus postvaccination surveillance data (9). Economic parameters (such as medical costs and quality-adjusted life-year [QALY] weights) were derived from published, U.S.-specific studies (9). All costs were inflated to 2021 U.S. dollars using the medical care component of the U.S. Consumer Price Index (10). In our base case, we assumed that the price of the 9-valent vaccine was $262 per dose (including administration fees) (Supplement Table 1, available at Annals.org).

Because HPV-ADVISE is an individual-based model, it allows the specific modeling of vaccination scenarios within subgroups of the modeled population based on individual risk factors, such as sexual behavior. The comparator scenario is based on the existing 9-valent HPV vaccination program of persons aged 9 to 26 years in the United States. In our base-case analysis, we examined the expansion of this program to include the vaccination of all mid-adults or to selectively include 2 specific subgroups of this population that may be at higher risk for HPV infection. These subgroups are mid-adults with higher sexual activity, and mid-adults who have just separated. On entry into the modeled population in HPV-ADVISE, individuals are attributed 1 of 4 levels of sexual activity based on the number of lifetime partners they are expected to have (from low [level L0] to high [level L3]) (Technical Appendix [8]). For the scenario involving vaccination of mid-adults with higher sexual activity, mid-adults from the higher levels L2 and L3 in the modeled population were vaccinated, corresponding to those who are expected to have more than 10 lifetime partners. For the scenario of vaccination of mid-adults who have just separated, vaccination occurred immediately on the termination of a stable partnership in the modeled population. This was possible because the HPV-ADVISE model tracks each individual's history, so we can know exactly when they end a stable partnership ("Partnership formation and separation process" in the Technical Appendix [8]).

HPV Vaccination Uptake

We modeled historical HPV vaccination uptake in the United States by age and year based on the same assumptions described by Laprise and colleagues (9) but included more recent HPV vaccination coverage data from the National Immunization Survey–Teen from 2017 to 2020 (11). In brief, we reproduced 2-dose and 3-dose coverage from data obtained from the National Immunization Survey–Teen for 13- to 17-year-olds and assumed that 18-year-olds have the same uptake rates as 17-year-olds. For vaccination coverage among young adults aged 19 through 26 years, we used the estimates from Chesson and colleagues (12) (see Supplement Figure 1 [available at Annals.org] for projected vaccination coverage of females and males over time). We assumed 100% vaccination uptake among unvaccinated mid-adults from 2021 onward to maximize the incremental benefits of mid-adult vaccination and reduce variability. For persons aged 9 to 17 years and for adults, we made the simplifying assumption that maximal vaccine efficacy occurs at the 2nd dose and 3rd dose, respectively. Vaccine efficacy is assumed to be prophylactic and to be 95% against HPV vaccine types with lifelong duration of protection.

Model Outcomes and Analysis

We did a cost-utility analysis from the health care payer perspective including only direct medical costs (see Supplement Table 4 [available at Annals.org] for the impact inventory for the health care perspective). The main outcomes were costs per QALY gained (incremental cost-effectiveness ratios [ICERs]). We discounted future costs and benefits at 3% per annum and used a 100-year time horizon from the start of HPV vaccination. Medical costs and QALYs were attributed to clinical outcomes over time. We also calculated NNV—the additional number of persons who would need to be vaccinated to prevent 1 additional HPV-related cancer case. In a sensitivity

Table 1. Cost-Effectiveness of Vaccinating Mid-adults in the Context of an Ongoing HPV Vaccination Program for Persons Aged 9–26 Years (Base-Case* Analysis)†

Vaccination Strategy	Median Additional Costs (80% UI), *$ (millions)*	Median QALYs Gained (80% UI), *n*	Median ICER (80% UI), *$/QALY gained*
Vaccinating all mid-adults	250.6 (249.1–251.9)	125.1 (19.4–204.5)	2 005 000 (1 227 000–>10 000 000)
Vaccinating mid-adults with higher sexual activity‡	86.9 (80.8–92.8)	98.5 (30.2–189.5)	763 000 (447 000–2 946 000)
Vaccinating mid-adults who have just separated	96.5 (94.0–127.1)	88.7 (21.2–193.8)	1 164 000 (525 000–4 627 000)

HPV = human papillomavirus; ICER = incremental cost-effectiveness ratio; QALY = quality-adjusted life-year; UI = uncertainty interval.
* Vaccinating persons aged 27–45 y. Cost per dose of 9-valent vaccine, $262 in persons aged 19–45 y. Discount rate, 3% per annum for future costs and benefits. Time horizon, 100 y from start of HPV vaccination.
† The comparator is vaccinating persons aged 9–26 y. Mid-adults are women and men aged 27–45 y. Model projections are presented as the median and 10th and 90th percentiles (80% UI) of results from the 50 best-fitting parameter sets identified through calibration. Open stable population of 1 million persons.
‡ Persons with higher sexual activity are those who are expected to have ≥10 lifetime partners (sexual activity levels L2 and L3; see the Technical Appendix [8]).

analysis, we used the individual-based feature of the HPV-ADVISE model to vary the vaccinated subgroups. These variations were based on individual risk factors integrated into HPV-ADVISE, including age, gender, sexual activity level, and cervical cancer screening behavior level (comprising 5 screening levels based on frequency, from annual to never screened) (Technical Appendix [8]). We also examined the effect of natural history assumptions (such as progression rates from infection to disease and higher incidence of vulvar, anal, and oropharyngeal cancer) and economic parameters (such as health care costs, disease burden [QALY weights and case fatality], vaccine price, discount rate, and time horizon). Model projections are presented using the median and 10th and 90th percentiles (80% uncertainty interval [UI]) of results from the 50 best-fitting parameter sets identified during model calibration, to consider variability due to uncertainty in sexual behavior, HPV infection, and progression of HPV-related diseases.

Role of the Funding Source

Coauthors from the Centers for Disease Control and Prevention contributed to the design of the study, interpretation of the findings, and editing of the manuscript.

Results

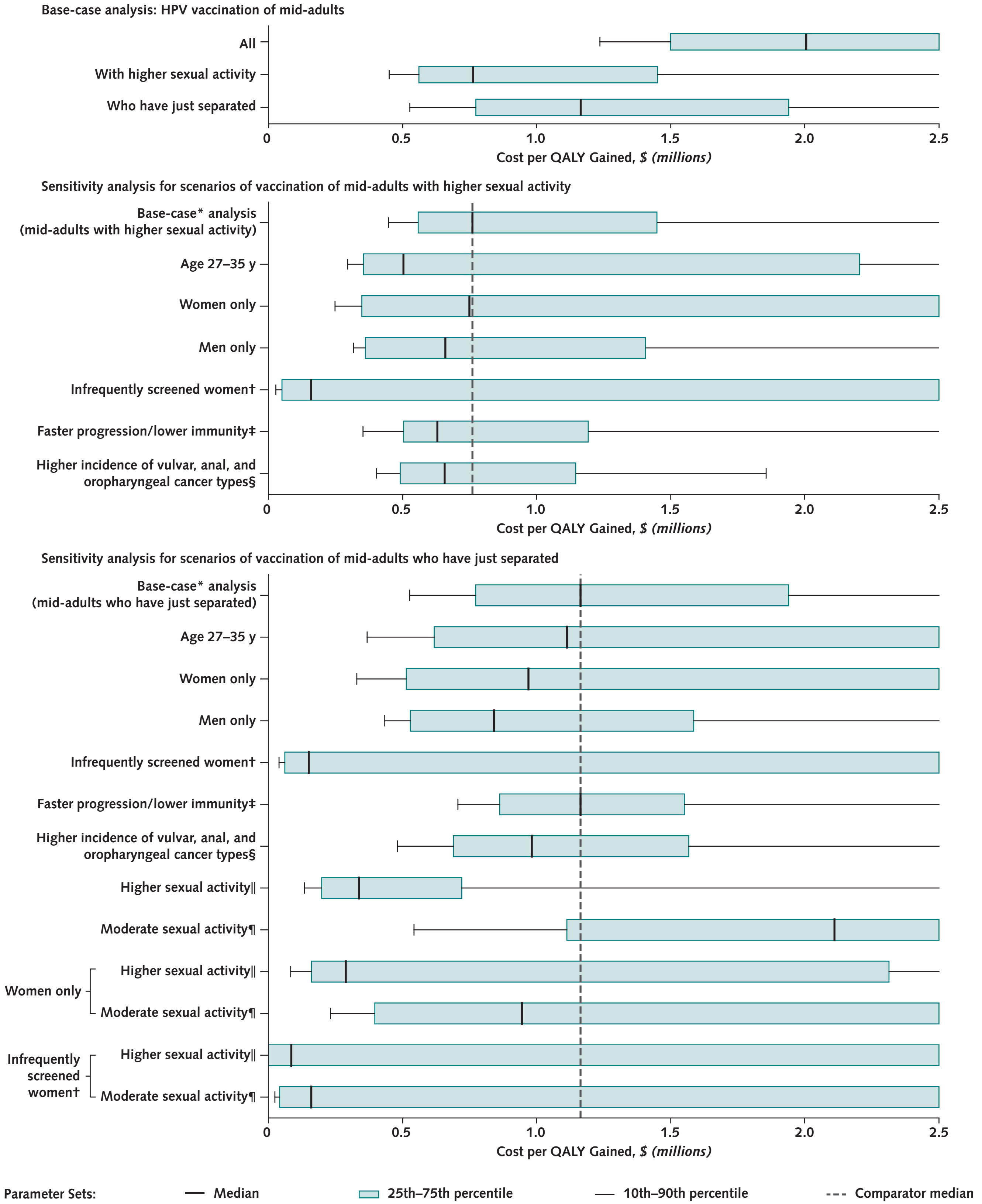
Under base-case assumptions, expanding an HPV vaccination program for persons aged 9 to 26 years to include all mid-adults is projected to cost $2 005 000 (80% UI, $1 227 000 to >$10 million) per QALY gained (**Table 1** and **Figure**, *top*). Mid-adult HPV vaccination is projected to cost $763 000 (80% UI, $447 000 to $2 946 000) per QALY gained when limited to persons with higher sexual activity and $1 164 000 (80% UI, $525 000 to $4 627 000) per QALY gained when limited to those who have just separated (**Table 1** and **Figure**, *top*). The model also projected NNVs of 7670 (80% UI, 5550 to 12 120), 3190 (80% UI, 2230 to 4750), and 5150 (80% UI, 3800 to 8250) to prevent 1 additional HPV-related cancer case when vaccinating all mid-adults and when limiting vaccination to mid-adults with higher sexual activity and those who have just separated, respectively (**Table 2**; **Supplement Table 2**, available at Annals.org). By comparison, 223 persons (80% UI, 210 to 232 persons) aged 9 to 26 years would need to be vaccinated to prevent 1 additional HPV-related cancer case (vs. no vaccination) (**Supplement Table 3**, available at Annals.org).

Results were most sensitive to level of sexual activity and screening behavior (**Tables 2** and 3; **Figure**, *middle* and *bottom*; **Supplement Table 2**; and **Supplement Figure 2**, *B* and *C*, available at Annals.org). Vaccinating infrequently screened (less frequently than every 5 years or never) mid-adult women who have higher sexual activity and have just separated produced the lowest ICER ($86 000 [80% UI, $3000 to >$10 million] per QALY gained) (**Table 3**) and the lowest NNV (470 [80% UI, 120 to >1 000 000] mid-adults vaccinated to prevent 1 additional cancer case) (**Table 2**; **Supplement Table 2**). Vaccinating only mid-adult men with higher sexual activity produced lower ICERs and NNVs than vaccinating only mid-adult women. Results were similar when vaccinating only men or women who have just separated (**Tables 2** and 3; **Figure**, *middle* and *bottom*; and **Supplement Table 2**). When economic parameters were varied, cost-effectiveness results were most sensitive to assumptions about the vaccine cost per dose, discount rate, and time horizon. Lower discount rates produced lower mid-adult vaccination ICERs, whereas shorter time horizons produced higher ICERs (**Table 3**). This is because the additional vaccination costs are mostly incurred during the first years of mid-adult vaccination while most of its benefits occur later (**Supplement Figure 3**, available at Annals.org).

Discussion

In this analysis, we examined the cost-effectiveness of expanding HPV vaccination to subgroups of the mid-adult U.S. population that may be at higher risk for acquiring HPV infection. Under all scenarios investigated, HPV vaccination of mid-adults was much less cost-effective than that of persons aged 26 years or younger. The estimated cost per QALY gained by

Figure. Cost-effectiveness of vaccinating mid-adults in the context of an ongoing vaccination program for persons aged 9–26 y.



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Mid-adults are defined as women and men aged 27–45 y. Box plots represent the median and 10th, 25th, 75th, and 90th percentiles of projections for the 50 best-fitting parameter sets. The dotted line represents the median for the comparator for the base case in sensitivity analyses. HPV = human papillomavirus; QALY = quality-adjusted life-year. **Top.** Base-case analysis. **Middle.** Sensitivity analysis for scenarios of vaccination of mid-adults with higher sexual activity (≥10 lifetime partners; sexual activity levels L2 and L3; see the Technical Appendix [8]). **Bottom.** Sensitivity analysis for scenarios of vaccination of mid-adults who have just separated.

* Vaccinating persons aged 27–45 y. Cost per dose of 9-valent vaccine, $262 in persons aged 19–45 y. Discount rate, 3% per annum for future costs and benefits. Time horizon, 100 y from start of HPV vaccination.

† Women screened less frequently than every 5 y or never being screened (screening behavior levels S3 and S4 in HPV-ADVISE; see the Technical Appendix [8]).

‡ We separated the 50 parameter sets into those that have lower probability of natural immunity after clearance in females (≤40% vs. >40%) and faster progression to cervical intraepithelial neoplasia (CIN) grade 1, 2, or 3 (e.g., average median time from infection to CIN1 of 10 vs. 15 mo and average median time from infection to CIN3 of 32 vs. 36 mo) (22/50 parameter sets) and those that have a higher natural immunity and slower progression (28/50 parameter sets).

§ We assumed a 50% increase in the incidence of vulvar, anal, and oropharyngeal cancer. This is illustrative and reflects the maximal scale of the observed increasing trends in HPV-related cancer incidence (13, 14). For this scenario, we also assumed that the incidence of vulvar, anal, and oropharyngeal cancer in the absence of vaccination is 50% higher than in the base case.

‖ Persons who are expected to have ≥10 lifetime partners (sexual activity levels L2 and L3; see the Technical Appendix [8]).

¶ Persons who are expected to have 2–9 lifetime partners (sexual activity level L1; see the Technical Appendix [8]).

vaccinating mid-adults was about $2 million for all mid-adults, $1.2 million for those who have just separated from a stable relationship, $0.7 million for those with higher sexual activity (adults who are expected to have ≥10 lifetime partners), and $86 000 for infrequently screened mid-adult women who have higher sexual activity and have just separated. In contrast, vaccinating persons aged 9 to 26 years is projected to be cost-saving (cost per QALY, <$0) compared with no vaccination (9). Similarly, the NNV to prevent 1 HPV-related cancer case was notably higher for mid-adults than for those aged 26 years or younger.

The current analysis projected fewer benefits per dose and higher ICERs from vaccinating mid-adults than our previous analysis (9) because of greater herd effects of HPV vaccination through age 26 years that reduce the additional benefits achievable by mid-adult vaccination. Three factors contribute to the greater projected herd effects compared with those in our previous analysis. First, our new estimates of historical vaccination coverage are based on data up to 2020 (**Supplement Figure 1**), which indicate higher coverage than assumed in the previous analysis based on 2016 data (9). Second, mid-adult vaccination starts in 2021 in the current analysis (vs. 2018 in the previous article), which provides 3 years of additional herd effects from the historical vaccination program. Most of the QALYs gained by extending vaccination to age 45 years come from vaccinating those up to age 35 years. An additional 3 years of herd effects can have a sizeable effect on HPV incidence in the group aged 27 to 35 years, which spans only 9 years. Third, in the current analysis, men aged 22 to 26 years are now included in the reference group to reflect the modified recommendation in 2019. Thus, the QALYs gained by vaccinating men aged 22 to 26 years are now included in the comparison strategy instead of the mid-adult strategy, and the additional herd effects from vaccinating men aged 22 to 26 years reduce the projected benefits of vaccinating mid-adults.

To our knowledge, our study is the first to examine the cost-effectiveness and NNV of vaccinating subgroups of mid-adults at higher risk for acquiring HPV infection. The study has 4 main strengths. First, our model (HPV-ADVISE) was calibrated to represent highly stratified, U.S.-specific sexual behavior, HPV epidemiology, and health care resource use. Second, HPV-ADVISE projections were shown to be consistent with age-specific data on the prevalence of postvaccination HPV infection and anogenital warts diagnosis from the United States (9) and Australia (15). Third, projections were made using 50 parameter sets to capture uncertainty in the natural history of HPV infection and related diseases and variability in sexual behavior data. Had we not used 50 parameter sets to account for uncertainty in the natural history of HPV, our UIs would not have been as wide and likely would not reflect the degree of uncertainty in the ICERs and NNVs of HPV vaccination of mid-adults. Finally, we performed extensive sensitivity analyses on key parameters and assumptions.

Our study also has limitations. First, although our model reproduces short-term observed herd effects after vaccination, the long-term herd effects of vaccinating younger age cohorts on mid-adult women and men remain uncertain. If HPV-ADVISE overestimates the herd effects of vaccination through age 26 years in the United States, the ICERs and NNVs of vaccinating mid-adult men and women might be more favorable than we estimated. Second, the natural history of HPV infections in mid-adults is not well understood. HPV-ADVISE assumes that progression from HPV infection to disease is independent of age. However, some have suggested that a proportion of redetection is due to reactivations of latent recurring infections (16) or deposition (17) and that new infections later in life have a smaller risk for progressing to cervical cancer (18–20); if this is true, mid-adult vaccination would produce lower benefits and higher ICERs and NNVs than projected. These benefits will also depend on the proportion of newly detected infections that are truly recent acquisitions and the potential for reactivated infections to produce HPV-related diseases. Third, historical vaccination coverage in the model could be underestimated because we assumed that maximal vaccine efficacy occurs at course completion. Recent immunogenicity and efficacy studies suggest that 1 dose of HPV vaccine might offer similar protection to 2 or 3 doses (21, 22). If so, this would result in a higher percentage of protection among persons aged 9 to 17 years than assumed, because observed coverage for 1 dose is higher than for 2 doses (23). However, higher historical

Table 2. Incremental NNV for Mid-adults to Prevent 1 Additional HPV-Related Cancer Case in the Context of an Ongoing HPV Vaccination Program for Persons Aged 9–26 Years*

Vaccination Strategy	Median Incremental NNV (80% UI)		
	Any HPV-Related Cancer†	**Female HPV-Related Cancer†**	**Male HPV-Related Cancer†**
Base-case‡ analysis			
Vaccinating all mid-adults	7670 (5550–12 120)	13 820 (8520–32 040)	18 550 (15 390–20 600)
Vaccinating mid-adults with higher sexual activity§	3190 (2230–4750)	5180 (3380–10 490)	8060 (7050–9390)
Vaccinating mid-adults who have just separated	5150 (3800–8250)	8440 (5150–16 410)	14 810 (12 270–16 440)
Sensitivity analyses			
Vaccinating mid-adults with higher sexual activity§			
Base case‡ (vaccinating mid-adults with higher sexual activity§)	3190 (2230–4750)	5180 (3380–10 490)	8060 (7050–9390)
Age			
Vaccinating persons aged 27–35 y	2700 (1530–5720)	4460 (2100–29 790)	6600 (5820–7800)
Gender			
Vaccinating mid-adult women only	3300 (1850–24 960)	4290 (2240–>1 000 000)	14 210 (8110–28 910)
Vaccinating mid-adult men only	2350 (1650–6210)	4190 (2460–21 910)	5440 (4370–7720)
Screening			
Vaccinating infrequently screened women‖ only	890 (320–>1 000 000)¶	1030 (320–>1 000 000)¶	15 510 (4580–>1 000 000)¶
Natural history			
Faster progression and lower immunity††	2770 (1500–3950)	4260 (2300–7500)	7860 (4060–9400)
Assuming 50% increase in incidence of vulvar, anal, and oropharyngeal cancer**	2530 (1810–3430)	4590 (3130–8890)	5480 (4790–6400)
Vaccinating mid-adults who have just separated			
Base case‡ (vaccinating mid-adults who have just separated)	5150 (3800–8250)	8440 (5150–16 410)	14 810 (12 270–16 440)
Age			
Vaccinating persons aged 27–35 y	4730 (3210–11 380)	7010 (4480–59 540)	13 050 (10 390–15 470)
Gender			
Vaccinating women only	4670 (2570–>1 000 000)	6740 (2990–>1 000 000)	23 530 (12 300–81 530)
Vaccinating men only	4220 (3020–9770)	6990 (4350–133 540)	10 390 (9170–11 870)
Screening			
Vaccinating infrequently screened women‖ only	2310 (510–>1 000 000)¶	2820 (530–>1 000 000)¶	48 400 (6650–>1 000 000)¶
Natural history			
Faster progression and lower immunity††	5130 (3720–7670)	8010 (5340–14 860)	14 630 (11 390–16 390)
Assuming 50% increase in incidence of vulvar, anal, and oropharyngeal cancer**	4150 (3200–6140)	7300 (4830–13 470)	10 060 (8360–11 180)
Level of sexual activity			
Women and men with moderate sexual activity‡‡	14 190 (6130–>1 000 000)	22 700 (7020–>1 000 000)¶	52 560 (30 500–108 980)
Women and men with higher sexual activity§	1730 (1040–3000)	2750 (1450–8160)	4130 (3200–7240)
Gender and level of sexual activity			
Moderate sexual activity‡‡, women only	6470 (2850–>1 000 000)¶	7290 (3150–>1 000 000)¶	63 240 (20 640–>1 000 000)¶
Higher sexual activity§, women only	1800 (750–>1 000 000)	2020 (880–>1 000 000)	9390 (4020–46 120)
Screening and level of sexual activity			
Moderate sexual activity‡‡, infrequently screened women‖ only	1390 (360–>1 000 000)¶	1470 (400–>1 000 000)¶	29 770 (3450–>1 000 000)¶
Higher sexual activity§, infrequently screened women‖ only	470 (120–>1 000 000)¶	640 (130–>1 000 000)¶	8850 (1080–>1 000 000)¶

HPV = human papillomavirus; NNV = number needed to vaccinate; UI = uncertainty interval.
* Mid-adults are women and men aged 27–45 y. Model projections are presented as the median and 10th and 90th percentiles (80% UI) of results from the best-fitting parameter sets identified through calibration. When mid-adult vaccination is restricted to women or men, benefits in the opposite gender are due entirely to indirect effects. See **Supplement Table 2** (available at Annals.org) for numbers of persons vaccinated and cancer cases averted.
† "HPV-related cancer" includes cancer of the cervix, vulva, vagina, anus, or oropharynx for females and cancer of the penis, anus, or oropharynx for males; it is defined as cancer at these sites caused by HPV, also sometimes called HPV-attributable cancer. "Any HPV-related cancer" includes cancer for both females and males.
‡ Vaccinating persons aged 27–45 y. Cost per dose of 9-valent vaccine, $262 in persons aged 19–45 y. Time horizon, 100 y from start of HPV vaccination.
§ Persons who are expected to have ≥10 lifetime partners (sexual activity levels L2 and L3; see the Technical Appendix [8]).
‖ Women screened less frequently than every 5 y or never being screened (screening behavior levels S3 and S4 in HPV-ADVISE; see the Technical Appendix [8]).
¶ Fewer than 75% of the parameter sets projected measurable additional HPV-related cancer cases averted for these scenarios because of stochasticity and small effect size.
** The value of 50% is illustrative and reflects the maximal scale of the observed increasing trends in HPV-related cancer incidence (13, 14). For this scenario, we also assume that the incidence of vulvar, anal, and oropharyngeal cancer in the absence of vaccination is 50% higher than in the base case.
†† We separated the 50 parameter sets into those that have lower probability of natural immunity after clearance in females (≤40% vs. >40%) and faster progression to cervical intraepithelial neoplasia (CIN) grade 1, 2, or 3 (e.g., average median time from infection to CIN1 of 10 vs. 15 mo and average median time from infection to CIN3 of 32 vs. 36 mo) (22/50 parameter sets) and those that have a higher natural immunity and slower progression (28/50 parameter sets).
‡‡ Persons who are expected to have 2–9 lifetime partners (sexual activity level L1; see the Technical Appendix [8]).

Table 3. Univariate Sensitivity Analysis: Cost-Effectiveness of Vaccinating Mid-adults in the Context of an Ongoing Vaccination Program for Persons Aged 9–26 Years*

Sensitivity Analysis	Median ICER (80% UI), *$ per QALY gained*	
	Vaccinating Mid-adults With Higher Sexual Activity†	**Vaccinating Mid-adults Who Have Just Separated**
Base case‡		
Vaccinating all mid-adults	763 000 (447 000–2 947 000)	1 164 000 (525 000–4 627 000)
Age		
Vaccinating persons aged 27–35 y	505 000 (292 000–>10 000 000)	1 113 000 (366 000–>10 000 000)§
Gender		
Vaccinating women only	751 000 (249 000–>10 000 000)	970 000 (326 000–>10 000 000)§
Vaccinating men only	662 000 (316 000–>10 000 000)	841 000 (432 000–>10 000 000)
Screening		
Vaccinating infrequently screened women‖ only	161 000 (29 000–>10 000 000)§	151 000 (40 000–>10 000 000)§
Natural history		
Faster progression and lower immunity¶	632 000 (355 000–2 895 000)	1 164 000 (707 000–3 566 000)
Assuming 50% increase in incidence of vulvar, anal, and oropharyngeal cancer**	659 000 (404 000–1 877 000)	982 000 (481 000–2 797 000)
Level of sexual activity		
Women and men with moderate sexual activity††	NA	2 110 000 (535 000–>10 000 000)§
Women and men with higher sexual activity†	NA	338 000 (132 000–>10 000 000)
Gender and level of sexual activity		
Moderate sexual activity††, women only	NA	945 000 (231 000–>10 000 000)§
Higher sexual activity†, women only	NA	289 000 (81 000–>10 000 000)
Screening and level of sexual activity		
Moderate sexual activity††, infrequently screened women‖ only	NA	160 000 (24 000–>10 000 000)§
Higher sexual activity†, infrequently screened women‖ only	NA	86 000 (3000–>10 000 000)§
Economic analysis parameters		
Vaccination cost per dose (base, $261.81)‡‡		
$168.98	484 000 (282 000–1 870 000)	739 000 (334 000–2 958 000)
$272.12	794 000 (466 000–3 065 000)	1 211 000 (547 000–4 812 000)
Maximum health care costs§§	752 000 (431 000–2 892 000)	1 145 000 (520 000–4 585 000)
Maximum disease burden§§	631 000 (362 000–2 024 000)	924 000 (453 000–3 211 000)
Discount rate		
0%	282 000 (186 000–563 000)	494 000 (342 000–883 000)
5%	1 363 000 (661 000–>10 000 000)	1 791 000 (702 000–>10 000 000)
50-y time horizon	1 256 000 (537 000–>10 000 000)	1 744 000 (604 000–>10 000 000)

ICER = incremental cost-effectiveness ratio; NA = not applicable; QALY = quality-adjusted life-year; UI = uncertainty interval.
* Mid-adults are women and men aged 27–45 y. Model projections are presented as the median and 10th and 90th percentiles (80% UI) of results from the best-fitting parameter sets identified through calibration.
† Persons who are expected to have ≥10 lifetime partners (sexual activity levels L2 and L3; see the Technical Appendix [8]).
‡ Vaccinating persons aged 27–45 y. Cost per dose of 9-valent vaccine, $262 in persons aged 19–45 y. Discount rate, 3% per annum for future costs and benefits. Time horizon, 100 y from start of vaccination.
§ Fewer than 75% of the parameter sets projected incremental QALY gains for these scenarios because of stochasticity and small effect size.
‖ Women screened less frequently than every 5 y or never being screened (screening behavior levels S3 and S4 in HPV-ADVISE; see the Technical Appendix [8]).
¶ We separated the 50 parameter sets into those that have lower probability of natural immunity after clearance in females (≤40% vs. >40%) and faster progression to cervical intraepithelial neoplasia (CIN) grade 1, 2, or 3 (e.g., average median time from infection to CIN1 of 10 vs. 15 mo and average median time from infection to CIN3 of 32 vs. 36 mo) (22/50 parameter sets) and those that have a higher natural immunity and slower progression (28/50 parameter sets).
** The value of 50% is illustrative and reflects the maximal scale of the observed increasing trends in HPV-related cancer incidence (13, 14). For this scenario, we also assume that the incidence of vulvar, anal, and oropharyngeal cancer in the absence of vaccination is 50% higher than in the base case.
†† Persons who are expected to have 2–9 lifetime partners (sexual activity level L1; see the Technical Appendix [8]).
‡‡ Vaccination costs are based on the Centers for Disease Control and Prevention vaccine price list as of 1 August 2018 (www.cdc.gov/vaccines/programs/vfc/awardees/vaccine-management/price-list/index.html) and include the cost of vaccine administration.
§§ Maximum estimates from the U.S. literature (see Supplement Table 1 [available at Annals.org] for health care costs used and the HPV-ADVISE Technical Appendix [8] for detailed QALY weights and case fatality).

coverages would result in higher ICERs and NNVs due to larger herd effects in mid-adults by vaccinating younger age cohorts. On the other hand, if fewer than 3 doses were recommended for adults, vaccinating mid-adults with 1 or 2 doses instead of 3, as we assumed, would result in much lower ICERs (if a 1- or 2-dose schedule is noninferior to a 3-dose schedule). However, NNVs would not change. Fourth, surveillance data show an increasing trend in vulvar, anal, and oropharyngeal cancer types (13, 14). In the HPV-ADVISE model, cancer incidence is stable over time in the absence of vaccination. However, even when we assumed a 50% increase in the incidence of vulvar, anal, and oropharyngeal cancer in sensitivity analyses, mid-adult vaccination ICERs and NNVs remained relatively high (Table 3 and Figure, *middle* and *bottom*). Fifth, cervical cancer screening guidelines and practice have changed in the United States (24). More effective cervical cancer screening could result in higher ICERs and NNVs for mid-adult vaccination. Sixth, our conclusions should not be generalized to low- to middle-income countries or other countries with different sexual behavior, HIV prevalence, screening participation, and vaccination coverages from the United States. Finally, issues related to the disruption of routine vaccination and cervical cancer screening due to the COVID-19 pandemic were not taken into account.

In conclusion, our results suggest that the cost-effectiveness and NNV of mid-adult vaccination improve when limited to mid-adult subgroups at higher risk for acquiring HPV infection. However, HPV vaccination of mid-adults remains much less cost-effective and produces much higher NNVs than vaccinating females and males aged 26 years or younger under all scenarios investigated.

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Note: Dr. Brisson had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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Reproducible Research Statement: *Study protocol*: The article, Supplement, and Technical Appendix publicly available on Dr. Brisson's website (www.marc-brisson.net/HPVadvise-US.pdf) are designed to provide sufficient information for an interested reader to replicate the analysis. *Statistical code*: Not applicable. *Data set*: Not applicable; this study did not use an analytic data set but used simulated data generated via our model, HPV-ADVISE. Model outputs are available on reasonable request.

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